Sponsor Description

Imagine if CAT Inspect and CAT AI Assistant joined forces to create something entirely new - a next-generation AI powerhouse for inspections. Picture the best features of both, seamlessly fused into a single, game-changing app. Unleash your creativity: leverage any AI technology at your disposal to build the ultimate solution that not only integrates these two tools, but redefines what's possible in field operations. This is your chance to set a new standard—think big, innovate boldly, and show us what the future of intelligent inspections looks like! In other words, if CAT Inspect and CAT AI Assistant have a baby!

The Challenge

Create a generative AI to revolutionize the way to inspect your machine!

Build an AI-powered tool that can automatically generate and evaluate multiple layout, and logistics plans for construction or industrial sites.

Goal

Use AI to optimize safety, efficiency, and cost during the inspection process —minimizing risky interactions, reducing travel and congestion, and streamlining equipment placement to cut operational overhead.

AI for Real-World Impact

Teams should explore how AI can transform field inspection processes. Enable inspectors to use voice and image inputs to instantly create structured, actionable reports - eliminating manual paperwork and delays. Participants are encouraged to reimagine this workflow, using AI to turn unstructured data into high-quality insights that accelerate decision-making and improve documentation accuracy. The solution should help operators reduce errors, improve safety, and accelerate inspections or site operations. Think beyond traditional analytics: design an agent that reasons over rich multimodal inputs to deliver actionable insights and drive real-world efficiency at scale.

Visual Parts Identification:

Another challenge is to use AI for visual parts identification and fitment. The task is to create a solution where users can snap a photo or describe a part, and the system returns ranked part numbers with fitment certainty, tailored to specific equipment models. This addresses a major pain point in support and logistics, reducing errors and improving customer experience.

Sample Videos for Daily Walkaround Inspection

[Cat® Wheel Loader | Daily Walkaround Inspection]

[Cat® Excavator Daily Walkaround Inspection]

Resources

- Access to LLMs (OpenAI/Gemini credits) or use cutting edge SOTA OSS Models from HF
- Templates and sample inspection prompts -> We will be providing a sample template for one make & model.
- Example images for PASS/FAIL/MONITOR scenarios -> We will be explaining the color codes used in the inspection template.
- Videos and guides on current inspection workflows
- Sample truck models and inspection standards

Judging Criteria

Projects will be evaluated on innovation, technical execution, impact, design, and presentation. The challenge is not about perfect code or polished decks, but about creative, functional, and impactful use of AI to solve real problems.

Mission

Use your creativity to build an AI solution that pushes the boundaries of what's possible in field operations and logistics. Show how AI can make inspections, parts identification, and site planning smarter, faster, and safer.